Indukuru Naveen Kumar Reddy

⚡ PYTHON DEVELOPER · UI/UX DESIGNER · AUTOMATION & MANUAL TESTING

MCA (2024–2026) · Open to Opportunities

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Bangalore, India

OBJECTIVE

Motivated and detail-oriented **MCA post-graduate (2024–2026)** seeking an entry-level position as a **Python Developer** or **Software Test Engineer**. Developed an innovative **Pupil Heart Rate Variability (HRV) Monitoring** system leveraging computer vision and pupillary fluctuation analysis, combining biomedical signal processing with machine learning to enable non-contact physiological monitoring. Possesses solid knowledge of **Python programming, Manual Testing, SDLC & STLC, Test Case Design, Defect Tracking, SQL, Computer Vision, and Generative AI**. Dedicated to ensuring software reliability, advancing biomedical AI research, and contributing to organizational success through analytical precision and continuous learning.

SKILLS & TECHNOLOGIES

PROGRAMMING LANGUAGES	Python, C, Java
FRONTEND TECHNOLOGIES	HTML5, CSS3, JavaScript, UI/UX
DATABASE	SQL, MongoDB (CRUD)
TESTING & QA	Manual Testing, SDLC / STLC, Test Case Design, Defect Tracking, Selenium
AI / ML / CV	OpenCV, TensorFlow, Scikit-Learn, Pandas, NumPy, Matplotlib, Generative AI
TOOLS & PLATFORMS	Git & GitHub, VS Code, Google Colab, Flask, Power BI
BIOMEDICAL RESEARCH	Pupil HRV Analysis, Pupillometry, Signal Processing, Real-time Detection

EDUCATION

Vikrama Simhapuri University — 2024 – 2026
MCA — Master of Computer Applications
Post-Graduate · Nellore, Andhra Pradesh
Pursuing

Viswam Degree & PG College — 2021 – 2024
BCA — Bachelor of Computer Applications
Under-Graduate
78%

Government Junior College — 2019 – 2021
Intermediate — C&E Vocational Course
Class XI & XII
81.8%

Government High School — 2018 – 2019
SSC — Secondary School Certificate
Class X Board Examination
98%

PROJECTS

Pupil Heart Rate Variability Monitoring via Pupillary Fluctuations

FEATURED PROJECT

Developed a novel non-contact biomedical system that monitors **Heart Rate Variability (HRV)** through real-time analysis of pupillary fluctuations captured via a standard webcam. The autonomic nervous system modulates pupil diameter in sync with cardiac cycles, enabling HRV estimation without physical sensors — bridging computer vision, biomedical signal processing, and machine learning.

- Implemented real-time pupil detection and segmentation using **OpenCV** and **Dlib** facial landmark detection with sub-pixel precision.
- Extracted pupillary oscillation time-series signals and applied **FFT-based frequency analysis** to isolate HRV-correlated bands (LF: 0.04–0.15 Hz, HF: 0.15–0.4 Hz).
- Computed standard HRV metrics — **SDNN, RMSSD, pNN50, LF/HF ratio** — from pupillary fluctuation data, validated against contact-based reference sensors.
- Developed a **Flask web application** with live video feed, real-time HRV dashboard, and autonomic stress-state classification using trained ML models.
- Applied Butterworth bandpass filtering, noise reduction, and artifact rejection algorithms to ensure signal fidelity under varied lighting conditions.
- Evaluated system accuracy against standard PPG reference, demonstrating reliable non-invasive autonomic biomarker estimation.

Python | OpenCV | Dlib | NumPy | SciPy | Scikit-Learn | Flask | Matplotlib | Signal Processing | HRV Analysis

Online Shopping Management System

FULL-STACK PROJECT

Built a comprehensive e-commerce platform with end-to-end user flow from authentication and product discovery to cart management and order tracking, using a modern full-stack architecture.

- Implemented **JWT-based user authentication/registration**, product catalog with search & filters, shopping cart, order placement, and payment gateway integration.
- Developed admin dashboard for inventory management, analytics, and reporting modules with Swagger docs and JUnit tests.
- Built REST APIs using **Spring Boot** with Spring Security + MySQL/PostgreSQL for backend; React + TypeScript + Tailwind for frontend.
- Implemented Redis for cart caching, product/category CRUD, checkout flow, order tracking, and admin panel for reports.

React | TypeScript | Spring Boot | Java | MySQL | PostgreSQL | Redis | JWT | Swagger | JUnit

INTERNSHIP

Classification of Iris Prediction

INTERNSHIP PROJECT

Supreme Infosolusions Private Limited · Python Full Stack Virtual Internship

Duration: 14 Weeks | **Period:** Dec 2025 - Apr 2026 | **Environment:** Windows, Visual Studio | **Languages:** Python, Flask, HTML, CSS

Worked on Iris Flower Classification using **SVM Machine Learning Algorithm** to predict species based on petal and sepal measurements.

- Applied data preprocessing, feature selection, and model training using **Support Vector Machine (SVM)** algorithm.
- Evaluated model performance using accuracy, precision, and recall, achieving high classification accuracy on the Iris dataset.
- Implemented the project in Python using libraries like **Pandas, NumPy, Matplotlib, and Scikit-Learn**.
- Deployed trained model via a **Flask web application** for real-time species classification through a clean UI.

Python | Scikit-Learn | SVM | Flask | Pandas | NumPy | Matplotlib | HTML | CSS

CERTIFICATIONS

Full Stack Data Science with Gen AI
Python Life — Kiran Sagar
Sep 23, 2026

Python Full Stack Virtual Internship
Supreme Infosolusions Pvt. Ltd.
Dec 2025 – Apr 2026

CRUD Operations in MongoDB
MongoDB, Inc.
Aug 21, 2025

Generative AI Mastermind
Outskill
2025

Career Guidance Session — Webinar
Grad Guru Innovation (ISO-certified)
Feb 17, 2026

Machine Learning Completion
Code Alpha
2025

SQL & Power BI Completion
Besant Technologies
2025

Python Completion Certificate
Kaggle
2025

Live Meeting Summarizer Application
Infosys
2025

ACADEMIC ACHIEVEMENTS

Central Zone Wicket-Keeper Batsman
Represented the central zone in inter-zone cricket as a specialised wicket-keeper batsman.

All India Softball Player
Selected and represented the national team at All India level softball competitions, demonstrating elite athletic discipline.

Taekwondo Silver Medallist
Earned Silver Medal in Taekwondo at State/National level competitions — displaying focus, resilience, and competitive spirit.

HOBBIES & INTERESTS

Playing Sports | Listening to Music | Exploring AI & ML | Biomedical Research | Open Source Coding